

Gibbs Sampling (Markov Chain Monte Carlo)

Want to simulate/generate random vectors from a distn π (a probability on $\mathbb{R}^k$)

We cannot invert cdf.

eg. $k = 2$.

$$F(a, b) = F(c, d) \quad \text{for } (a, b) \neq (c, d)$$

Inverse does not exist.

$$U \sim \text{Unif}(0, 1)$$

Multivariate normal
Multinomial

$$\pi(x_1, \dots, x_k)$$

$$x_0 = \begin{pmatrix} x_{10} \\ x_{20} \end{pmatrix}$$

For $x_1, \dots, x_k$:

$$X_1 \sim \pi_1(\cdot \mid x_2, \dots, x_k) \quad \text{univariate inverse cdf}$$

$$X_2 \sim \pi_2(\cdot \mid x_1, x_3, \dots, x_k)$$

$$\vdots$$

$$X_k \sim \pi_k(\cdot \mid x_1, \dots, x_{k-1})$$

$$x_1, \dots, x_n \sim \pi$$

$$\text{eg. } \pi = N_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & p \\ p & 1 \end{pmatrix}\right),$$

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \pi$$

$$X \mid Y = y \sim \mathcal{N}(py, 1 - p^2)$$

$$Y \mid X = x \sim \mathcal{N}(px, 1 - p^2)$$

$$-\frac{1}{2} \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & p \\ p & 1 \end{pmatrix}^{-1} \begin{pmatrix} x \\ y \end{pmatrix} + \frac{1}{2} y^2 \frac{1}{1 - p^2}$$

$$z_0 = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

First step:

$$x_1 \sim \mathcal{N}(py_0, \sqrt{1-p^2})$$

$$y_1 \sim \mathcal{N}(px_1, \sqrt{1-p^2})$$

$$x_2 \sim \mathcal{N}(py_1, \sqrt{1-p^2})$$

$$y_2 \sim \mathcal{N}(px_2, \sqrt{1-p^2})$$

eg2. 1-dimensional Ising model

$$z = \{x_1, \dots, x_d\} \quad \text{T temp} \quad d \text{ particles}$$

Each particle has a characteristic

$$x_i \in \{+1, -1\} \quad (\text{by spin})$$

$$\pi(x) \propto \exp\left(\frac{1}{T} \sum_{i=1}^{d-1} x_i x_{i+1}\right)$$

How to generate configuration x from the distn π .

$$\theta = \mathbb{E}[h(X)]$$

$$\pi(x) = c \exp\left(\frac{1}{T} \sum_{i=1}^{d-1} x_i x_{i+1}\right)$$

$$\pi_2(x_2 \mid x_{(-2)}) = \frac{\exp\left(\frac{1}{T} \sum_{i=1}^{d-1} x_i x_{i+1}\right)}{\exp(\dots)|_{x_2=+1} + \exp(\dots)|_{x_2=-1}}$$

$$= \frac{\exp\left(\frac{1}{T}(x_1 + x_3)\right)}{\exp\left(\frac{1}{T}(x_1 + x_3)\right) + \exp\left(-\frac{1}{T}(x_1 + x_3)\right)}$$

$$\pi_i(x_i = 1 \mid x_{(-i)}) = \frac{e^{\frac{1}{T}(x_{i-1} + x_{i+1})}}{e^{\frac{1}{T}(x_{i-1} + x_{i+1})} + e^{-\frac{1}{T}(x_{i-1} + x_{i+1})}}$$

$$\pi_i(x_i = -1 \mid x_{(-i)}) = p_i$$

Start with some configuration. Run the chain.

At nth step:

$$\pi_i(x_i = -1 \mid x_{(-i)}) = p_i.$$

Generate $U \sim U(0, 1)$. If $U < p_i$, then $x_i = -1$, else $x_i = +1$.